JOINS OF RELATIVIZED NECESSITIES ON A POSET: AN ALTERNATING-CHAIN CRITERION AND THE INDUCTION SCHEME OF LAX LOGIC

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ABSTRACT. For a poset P and a subset $C \subseteq P$, the relativized necessity $\bigcirc_C U = \{x : \uparrow x \cap C \subseteq U\}$ is a nucleus on the Heyting algebra of up-sets of P; on a finite poset every nucleus has this form, meets of nuclei correspond to unions of subsets and joins to intersections. We describe the join of two relativized necessities on an arbitrary poset as a greatest fixed point, $(\bigcirc_C \vee \bigcirc_{C'})(U) = P \setminus X$ with X the largest subset of $P \setminus U$ in which every point lies below a point of $X \cap C$ and below a point of $X \cap C'$, and we prove that the join equals the necessity $\bigcirc_{C \cap C'}$ relativized to the intersection if and only if P contains no infinite chain $q_1 < q_2 < \cdots$ alternating between $C \setminus C'$ and $C' \setminus C$ and avoiding $C \cap C'$ on every interval $[q_1, q_i]$. Consequently, when every nonempty subset of P has a maximal element, $C \mapsto \bigcirc_C$ turns arbitrary unions into meets and arbitrary intersections into joins, and by Lindenhovius's classification it is then an isomorphism from the subsets of P, ordered by reverse inclusion, onto the lattice of nuclei; for every poset with an infinite ascending chain some binary join is not preserved. Read as a frame condition, the criterion is exactly the validity of the induction scheme of the trimodal lax logic of two nuclei and their join on the frame (P, C, C') with the join interpreted as $\bigcirc_{C \cap C'}$; the condition is not first-order definable, as for the induction axiom of common knowledge, whose role the join plays. Finally, since finite meets of open and closed nuclei on $\mathrm{Up}(P)$ are relativized necessities, every nucleus on $\mathrm{Up}(P)$ is a join of relativized necessities, although the double negation nucleus on the binary tree is not a join of finitely many.

1. INTRODUCTION

Let P be a poset and $\mathrm{Up}(P)$ the Heyting algebra of its up-sets. For a subset $C \subseteq P$ the operator

$$\bigcirc_C U = \{x \in P : \uparrow x \cap C \subseteq U\},$$

"U holds at every point of C above x", is a nucleus on $\mathrm{Up}(P)$: it is inflationary, idempotent and preserves intersections. We call it the *necessity relativized to* C. Under the dictionary between nuclei on $\mathrm{Up}(P)$ and

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Grothendieck topologies on the dual poset, $\bigcirc_C$ is the topology generated by the subset C in the sense of Lindenhovius [6], who proved that every nucleus on $\mathrm{Up}(P)$ has this form if and only if every nonempty subset of P has a maximal element; in particular on a finite poset, where the assignment $C \mapsto \bigcirc_C$ turns unions into meets and intersections into joins in the lattice $\mathrm{N}(\mathrm{Up}(P))$ of nuclei [8, Thm. 4.1]. This paper is about joins on an arbitrary poset. Meets behave as before, $\bigcirc_C \wedge \bigcirc_{C'} = \bigcirc_{C\cup C'}$ pointwise, for arbitrary families. Joins do not: the join of two nuclei is the least nucleus above both, whose fixed points are the common fixed points, and the following theorem says exactly when it is again a relativized necessity.

Theorem 1.1. *Let P be a poset, $C, C' \subseteq P$ and $D = C \cap C'$. Then $\bigcirc_C \vee \bigcirc_{C'} \leq \bigcirc_D$ in $\mathrm{N}(\mathrm{Up}(P))$, with equality if and only if there is no infinite chain $q_1 < q_2 < q_3 < \cdots$ in P with $q_i \in C \setminus C'$ for odd i and $q_i \in C' \setminus C$ for even i such that no point of D lies in any interval $[q_1, q_i]$.*

We call such a chain a *D-free alternating chain*. When it exists the join is described instead by a greatest fixed point: for every up-set U,

$$(\bigcirc_C \vee \bigcirc_{C'})(U) = P \setminus X_U, \qquad X_U = \text{the largest } X \subseteq P\setminus U \text{ with } X \subseteq {\downarrow}(X \cap C) \cap {\downarrow}(X \cap C')$$

(Proposition 2.3), the lax analogue of the greatest-fixed-point definition of common knowledge, with $\bigcirc_C$ and $\bigcirc_{C'}$ in the role of the knowledge of the two agents. On a finite poset the greatest fixed point is reached after $|P|$ steps and X_U is empty exactly when it should be, so that the join is $\bigcirc_{C\cap C'}$ and equals the iterate $(\bigcirc_C \bigcirc_{C'})^{|P|}$ of [8]; on the chain $\omega + 2$ with C the even and C' the odd ordinals, the chain $0 < 1 < 2 < \cdots$ is $\varnothing$-free, and the join sends $\varnothing$ to $\uparrow\omega$ while $\bigcirc_\varnothing$ is the constant $\top$.

The theorem has a modal reading. In [8] the trimodal lax logic **LJ** of two nuclei $\bigcirc_1, \bigcirc_2$ and their join $\bigcirc$ was axiomatized by the lax axioms, the inclusions $\bigcirc_1\varphi \to \bigcirc\varphi$, $\bigcirc_2\varphi \to \bigcirc\varphi$ and the induction scheme

$$\text{(JL)} \qquad (\bigcirc_1\psi \to \psi) \to (\bigcirc_2\psi \to \psi) \to (\varphi \to \psi) \to (\bigcirc\varphi \to \psi),$$

and it was shown to have the finite model property with respect to the frames (P, C, C'), P a finite poset, with $\bigcirc_1 = \bigcirc_C$, $\bigcirc_2 = \bigcirc_{C'}$ and $\bigcirc = \bigcirc_{C\cap C'}$. On an arbitrary poset these frames are models of the bimodal fragment, and $\bigcirc_{C\cap C'}$ is the natural candidate for the third connective. Theorem 1.1 says that (JL) is valid on (P, C, C') under this reading if and only if P has no D-free alternating chain (Theorem 4.1); and the class of such frames is not first-order definable (Proposition 4.2), as the induction axiom of common knowledge defines a non-elementary class. The join itself is always sound when interpreted by the greatest fixed point.

What is known. Nuclei on Heyting algebras and their lattice are classical [7, 5]; the join of two nuclei is the least common fixed point operator, computed by transfinite iteration of the composite [7, 1]. The nuclei $\bigcirc_C$ are the "relativized" or "subframe" modalities of intuitionistic Kripke semantics and are the Lawvere–Tierney topologies of the presheaf topos on P [2, 8]; as

Grothendieck topologies on the dual poset they are the topologies J_X generated by a subset, and Lindenhovius [6, Prop. 2.11, Thm. 2.12] showed that $X \mapsto J_X$ is an order isomorphism from the subsets onto all Grothendieck topologies exactly for Artinian posets, i.e. in our orientation when every nonempty subset of P has a maximal element; Hemelaer [3] describes the topologies with enough points and of finite type on general posets. The finite case of Theorem 1.1, the iterate formula and the logic **LJ** are from [8]. Common knowledge as a greatest fixed point and its induction axiom are from [4].

What is new. The two main results are the alternating-chain criterion for the join of two relativized necessities on an arbitrary poset (Theorem 1.1, with its version for countable families in Theorem 3.1) and the frame correspondence for the induction scheme (Theorem 4.1). The greatest-fixed-point description of the join (Proposition 2.3), the preservation of arbitrary joins when every nonempty subset has a maximal element (Corollary 3.2, which with Lindenhovius's classification makes $C \mapsto \bigcirc_C$ an isomorphism of complete lattices there) and the non-definability of the frame condition (Proposition 4.2) are supporting results. Theorem 5.1, that every nucleus on $\mathrm{Up}(P)$ is a join of relativized necessities, is the classical generation of the lattice of nuclei by open and closed nuclei [5] together with the observation that on $\mathrm{Up}(P)$ the finite meets of open and closed nuclei are relativized necessities; the binary-tree example showing that finitely many do not suffice is new.

2. Relativized necessities

Throughout, P is a poset, $\mathrm{Up}(P)$ its complete Heyting algebra of up-sets, and for $S \subseteq P$ we write $\downarrow S$ and $\uparrow S$ for the down-set and the up-set generated by S. A *nucleus* on a Heyting algebra H is a map $j\colon H \to H$ with $a \le ja$, $jja \le ja$ and $j(a \wedge b) = ja \wedge jb$; its fixed points $\mathrm{Fix}(j) = \{a : ja = a\}$ are closed under meets and under $b \mapsto a \to b$, and j is recovered from $\mathrm{Fix}(j)$ as $ja = \bigwedge\{b \in \mathrm{Fix}(j) : b \ge a\}$ when these meets exist. The nuclei of H form a complete lattice $\mathrm{N}(H)$ under the pointwise order, with pointwise meets; $j \le k$ if and only if $\mathrm{Fix}(k) \subseteq \mathrm{Fix}(j)$, and when H is complete the join $j \vee k$ is the nucleus with $\mathrm{Fix}(j \vee k) = \mathrm{Fix}(j) \cap \mathrm{Fix}(k)$, i.e. $(j \vee k)(a)$ is the least common fixed point above a [7, 1].

Definition 2.1. For $C \subseteq P$ and $U \in \mathrm{Up}(P)$ put $\bigcirc_C U = \{x \in P : \uparrow x \cap C \subseteq U\} = P \setminus \downarrow(C \setminus U)$.

Lemma 2.2. (i) *$\bigcirc_C$ is a nucleus on $\mathrm{Up}(P)$, and $\mathrm{Fix}(\bigcirc_C) = \{U \in \mathrm{Up}(P) : P \setminus U \subseteq \downarrow(C \setminus U)\}$.*

(ii) *$C \mapsto \bigcirc_C$ is injective and order-reversing, and for every family $(C_i)_{i\in I}$, $\bigcirc_{\bigcup_i C_i} = \bigwedge_i \bigcirc_{C_i}$.*

(iii) *$\bigcirc_C \vee \bigcirc_{C'} \le \bigcirc_{C\cap C'}$.*

Proof. (i) $\bigcirc_C U$ is an up-set containing U, $\bigcirc_C$ preserves intersections, and if $x \in \bigcirc_C \bigcirc_C U$ and $c \in \uparrow x \cap C$ then $c \in \bigcirc_C U$, so $c \in \uparrow c \cap C \subseteq U$. The description of the fixed points is the definition. (ii) If $c \in C \setminus C'$, then $c \notin \bigcirc_C(P \setminus \downarrow c)$ while $c \in \bigcirc_{C'}(P \setminus \downarrow c)$, because the points of C' above c are different from c, hence not below c. Reversal of order and the formula for unions are pointwise: $\uparrow x \cap \bigcup_i C_i \subseteq U$ iff $\uparrow x \cap C_i \subseteq U$ for all i, and meets of nuclei are pointwise. (iii) is (ii) for the inclusions $C \cap C' \subseteq C, C'$. □

Proposition 2.3 (the join as a greatest fixed point)**.** *For $U \in \mathrm{Up}(P)$ let X_U be the union of all $X \subseteq P \setminus U$ with $X \subseteq \downarrow(X \cap C) \cap \downarrow(X \cap C')$. Then X_U is a down-set with the same property, and $(\bigcirc_C \vee \bigcirc_{C'})(U) = P \setminus X_U$.*

Proof. A union of sets with the property has the property, so X_U has it, and $\downarrow X_U \subseteq P \setminus U$ has it too, so X_U is a down-set. Put $V = P \setminus X_U$, an up-set containing U; it is fixed by $\bigcirc_C$ by Lemma 2.2(i), because $P \setminus V = X_U \subseteq \downarrow(X_U \cap C) = \downarrow(C \setminus V)$, and likewise by $\bigcirc_{C'}$. Conversely, if $W \supseteq U$ is fixed by both, then $X = P \setminus W$ satisfies $X \subseteq \downarrow(C \setminus W) = \downarrow(X \cap C)$ and the same for C', so $X \subseteq X_U$ and $W \supseteq V$. Hence V is the least common fixed point above U. □

On a finite poset, where every nucleus is a relativized necessity [6, Thm. 2.12], [8, Thm. 4.1], X_U is reached by iterating $X \mapsto (P \setminus U) \cap \downarrow(X \cap C) \cap \downarrow(X \cap C')$ from $P \setminus U$, which removes at least one point per step; this is the iterate formula $\bigcirc_C \vee \bigcirc_{C'} = (\bigcirc_C \bigcirc_{C'})^{|P|}$ of [8, Thm. 4.1] in complemented form.

3. The alternating-chain criterion

Proof of Theorem 1.1. Write $D = C \cap C'$ and $j = \bigcirc_C \vee \bigcirc_{C'}$. The inequality $j \leq \bigcirc_D$ is Lemma 2.2(iii), and equality means $\mathrm{Fix}(\bigcirc_C) \cap \mathrm{Fix}(\bigcirc_{C'}) \subseteq \mathrm{Fix}(\bigcirc_D)$, since $\mathrm{Fix}(j) = \mathrm{Fix}(\bigcirc_C) \cap \mathrm{Fix}(\bigcirc_{C'})$ and the reverse inclusion is $j \leq \bigcirc_D$.

Suppose there is no D-free alternating chain, let U be fixed by $\bigcirc_C$ and $\bigcirc_{C'}$, and let $p \notin U$. We must find a point of $D \setminus U$ above p. Suppose there is none. Since $U \in \mathrm{Fix}(\bigcirc_C)$ there is $q_1 \in C \setminus U$ above p, and $q_1 \notin D$ by assumption, so $q_1 \in C \setminus C'$. Since $U \in \mathrm{Fix}(\bigcirc_{C'})$ there is $q_2 \in C' \setminus U$ above q_1; again $q_2 \notin D$, so $q_2 \in C' \setminus C$, and $q_2 \neq q_1$ because $C \setminus C'$ and $C' \setminus C$ are disjoint, i.e. $q_1 < q_2$. Continuing, we obtain an infinite alternating chain $q_1 < q_2 < \cdots$ in $P \setminus U$, strictly increasing for the same reason at every step. A point $d \in D$ in some interval $[q_1, q_i]$ would be below $q_i \notin U$, hence outside the up-set U, and above p, contrary to the assumption; so the chain is D-free, a contradiction. Hence $P \setminus U \subseteq \downarrow(D \setminus U)$, i.e. $U \in \mathrm{Fix}(\bigcirc_D)$.

Conversely, let $q_1 < q_2 < \cdots$ be a D-free alternating chain and put $U = P \setminus \downarrow\{q_i : i \geq 1\}$, an up-set. It is fixed by $\bigcirc_C$: a point $x \notin U$ lies below some q_i, hence below q_j for every odd $j \geq i$, and $q_j \in C \setminus U$. Symmetrically U is fixed by $\bigcirc_{C'}$. But $U \notin \mathrm{Fix}(\bigcirc_D)$: the point $q_1 \notin U$ is below no point of

$D \setminus U$, because such a point would lie in $\downarrow\{q_i\}$, i.e. in some interval $[q_1, q_i]$. So $\mathrm{Fix}(\bigcirc_C) \cap \mathrm{Fix}(\bigcirc_{C'}) \not\subseteq \mathrm{Fix}(\bigcirc_D)$ and $j \neq \bigcirc_D$. □

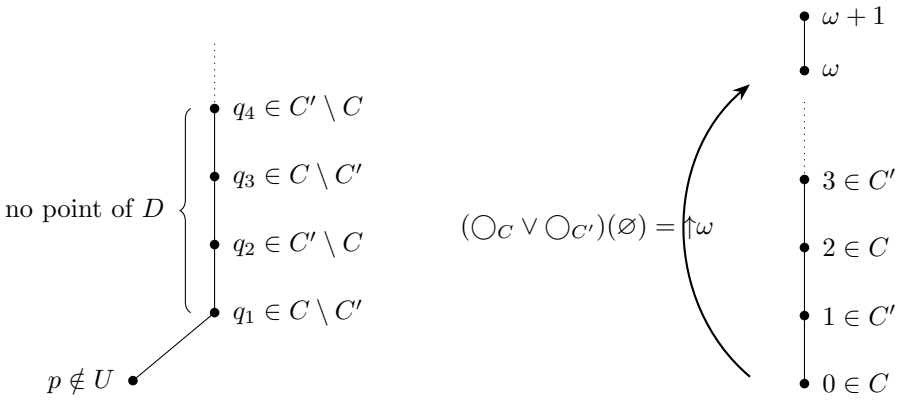


FIGURE 1. Left: a D-free alternating chain above a point $p \notin U$, the obstruction of Theorem 1.1. Right: the chain $\omega + 2$ with C the even and C' the odd finite ordinals; the alternating chain $0 < 1 < 2 < \cdots$ is $\varnothing$-free, and the join of the two nuclei sends $\varnothing$ to $\uparrow\omega$, whereas $\bigcirc_{C \cap C'} = \bigcirc_\varnothing$ is the constant P.

Theorem 3.1 (countable families)**.** *Let $(C_i)_{i \in I}$ be a nonempty countable family of subsets of P and $D = \bigcap_i C_i$. Then $\bigvee_i \bigcirc_{C_i} = \bigcirc_D$ if and only if there is no infinite chain $q_1 < q_2 < \cdots$ in $P \setminus D$ such that no point of D lies in any interval $[q_1, q_k]$ and every C_i contains q_k for infinitely many k.*

Proof. As before, $\bigvee_i \bigcirc_{C_i} = \bigcirc_D$ iff $\bigcap_i \mathrm{Fix}(\bigcirc_{C_i}) \subseteq \mathrm{Fix}(\bigcirc_D)$. Let U be a common fixed point and $p \notin U$ a point with no point of $D \setminus U$ above it. Choose a sequence of indices $i_1, i_2, \ldots$ in which every index of I occurs infinitely often. Put $q_0 = p$ and, given $q_{k-1} \notin U$, use $U \in \mathrm{Fix}(\bigcirc_{C_{i_k}})$ to choose $q_k \in C_{i_k} \setminus U$ with $q_k \geq q_{k-1}$; by assumption $q_k \notin D$. The sequence is non-decreasing and is not eventually constant: a value q taken for all large k would lie in C_i for every i, since every index recurs, i.e. in D. Being non-decreasing and not eventually constant, it takes each value only finitely many times. Its distinct values therefore form an infinite chain in $P \setminus U$ in which each C_i occurs infinitely often, because C_i contains q_k for the infinitely many k with $i_k = i$, and which is D-free, because a point of D below some value is outside the up-set U and above p. This contradicts the hypothesis, so $U \in \mathrm{Fix}(\bigcirc_D)$. Conversely, given a chain as in the statement, the up-set $U = P \setminus \downarrow\{q_k\}$ is fixed by every $\bigcirc_{C_i}$, since each C_i is cofinal in the chain, and $q_1 \notin U$ lies below no point of $D \setminus U$; so U is a common fixed point not fixed by $\bigcirc_D$, by the same computation. □

Corollary 3.2. *Suppose every nonempty subset of P has a maximal element (equivalently, assuming dependent choice, P has no infinite ascending chain). Then $\bigvee_{i\in I} \bigcirc_{C_i} = \bigcirc_{\bigcap_i C_i}$ for every family $(C_i)_{i\in I}$ of subsets of P, so that $C \mapsto \bigcirc_C$ turns arbitrary unions into meets and arbitrary intersections into joins; by Lindenhovius's theorem it is then an isomorphism of complete lattices $(\mathcal{P}(P), \supseteq) \cong \mathrm{N}(\mathrm{Up}(P))$. For every poset with an infinite ascending chain there are $C, C' \subseteq P$ with $\bigcirc_C \vee \bigcirc_{C'} \neq \bigcirc_{C\cap C'}$.*

Proof. Let $D = \bigcap_i C_i$ (with $D = P$ and $\bigcirc_D$ the identity if I is empty), let U be fixed by every $\bigcirc_{C_i}$ and let $p \notin U$. The set $\uparrow p \setminus U$ is nonempty, so it has a maximal element m. For each i, since $U \in \mathrm{Fix}(\bigcirc_{C_i})$ there is $c_i \in C_i \setminus U$ with $m \leq c_i$, and maximality gives $c_i = m$; hence $m \in D \setminus U$ lies above p. So $U \in \mathrm{Fix}(\bigcirc_D)$, and $\bigcap_i \mathrm{Fix}(\bigcirc_{C_i}) \subseteq \mathrm{Fix}(\bigcirc_D)$ gives the equality of nuclei. Preservation of unions is Lemma 2.2(ii), and injectivity makes $C \mapsto \bigcirc_C$ an order-embedding of complete lattices; it is onto $\mathrm{N}(\mathrm{Up}(P))$ by [6, Thm. 2.12] applied to the dual poset, since every nonempty subset of P has a maximal element iff every nonempty subset of the dual has a minimal element. For the last statement let $q_1 < q_2 < \cdots$ be an infinite chain and put $C = \{q_i : i \text{ odd}\}$, $C' = \{q_i : i \text{ even}\}$; then $D = \varnothing$ and the chain is D-free, so Theorem 1.1 applies. □

Example 3.3. (a) On $\omega + 2$ with C the even and C' the odd finite ordinals, the join sends $\varnothing$ to $\uparrow\omega$ and $\bigcirc_\varnothing$ is constant; this is the pair of nuclei used in [8, §5] to show that the join is not expressible in the bimodal language. (b) On $\omega + 1$ with $C = \{\text{even}\} \cup \{\omega\}$ and $C' = \{\text{odd}\} \cup \{\omega\}$, $D = \{\omega\}$ is above every finite ordinal but inside no interval $[q_1, q_i]$ of the chain $0 < 1 < 2 < \cdots$, so the join differs from $\bigcirc_D$: the up-set $\{\omega\}$ is fixed by $\bigcirc_C$ and $\bigcirc_{C'}$ but $\bigcirc_D\{\omega\} = P$. (c) On $\mathbb{N}$ with $C = \{n : n \equiv 0, 2, 3 \pmod 4\}$ and $C' = \{n : n \equiv 0, 1, 3 \pmod 4\}$ we have $C \setminus C' = \{n \equiv 2\}$, $C' \setminus C = \{n \equiv 1\}$, and every interval $[q_1, q_2]$ with $q_1 \equiv 2$, $q_2 \equiv 1$, $q_1 < q_2$ contains $q_1 + 1 \equiv 3 \in D$. Hence no D-free alternating chain exists although $\mathbb{N}$ has infinite chains, and $\bigcirc_C \vee \bigcirc_{C'} = \bigcirc_D$.

Remark 3.4. The pattern is that of the lattice of Grothendieck topologies inside the lattice of classes of local isomorphisms [9, 10]: there, meets of topologies always go to joins of classes, and joins of topologies go to meets of classes on finite categories but not on the poset $\mathbb{N}$. Here, $C \mapsto \bigcirc_C$ always turns unions into meets, and turns intersections into joins exactly when the alternating chains of Theorem 1.1 are absent, which is automatic without infinite ascending chains.

4. The induction scheme as a frame condition

A *frame* is a triple (P, C, C') with P a poset and $C, C' \subseteq P$. A valuation assigns up-sets to propositional variables; $\wedge, \vee, \to, \bot$ are interpreted in $\mathrm{Up}(P)$, $\bigcirc_1$ by $\bigcirc_C$ and $\bigcirc_2$ by $\bigcirc_{C'}$. For the third connective $\bigcirc$ there are two readings: the *intersection reading* $\bigcirc = \bigcirc_{C\cap C'}$ and the *join reading*

$\bigcirc = \bigcirc_C \vee \bigcirc_{C'}$, given by Proposition 2.3. The two coincide on finite frames [8, Thm. 4.1].

Theorem 4.1. (i) *Under the join reading every frame validates* **LJ**.
(ii) *Under the intersection reading a frame validates the inclusion axioms* $\bigcirc_1\varphi \to \bigcirc\varphi$ *and* $\bigcirc_2\varphi \to \bigcirc\varphi$ *and the lax axioms for* $\bigcirc$*, and it validates the induction scheme (JL) if and only if it has no* D*-free alternating chain,* $D = C \cap C'$*; in that case it validates* **LJ**.

Proof. (i) $\mathrm{Up}(P)$ is a complete Heyting algebra and $\bigcirc_C, \bigcirc_{C'}$ are nuclei, so $(\mathrm{Up}(P), \bigcirc_C, \bigcirc_{C'}, \bigcirc_C \vee \bigcirc_{C'})$ is an **LJ**-algebra by [8, Prop. 3.2]. (ii) $\bigcirc_{C\cap C'}$ is a nucleus above $\bigcirc_C$ and $\bigcirc_{C'}$, which gives the inclusion and lax axioms. If there is no D-free alternating chain, $\bigcirc_{C\cap C'}$ is the join by Theorem 1.1 and (i) applies. If there is one, then by Theorem 1.1 some up-set U is fixed by $\bigcirc_C$ and $\bigcirc_{C'}$ but not by $\bigcirc_D$; take a variable p with value U and the instance of (JL) with $\psi = \varphi = p$. Its antecedents $\bigcirc_1 p \to p$, $\bigcirc_2 p \to p$ and $p \to p$ have value P, while $\bigcirc p \to p$ has value $\bigcirc_D U \to U \neq P$ because $\bigcirc_D U \not\subseteq U$. So this instance of (JL) fails. □

Proposition 4.2. *The class of frames* (P, C, C') *without* D*-free alternating chains is not definable by any set of first-order sentences in the language of a binary relation and two unary predicates. Hence the induction scheme (JL), under the intersection reading, is not first-order definable as a frame condition.*

Proof. Suppose a set Σ of sentences defined the class. Add constants $c_1, c_2, \ldots$ and the sentences $c_i < c_{i+1}$, $c_i \in C \setminus C'$ for odd i, $c_i \in C' \setminus C$ for even i, and $\neg\exists d\,(d \in C \wedge d \in C' \wedge c_1 \leq d \leq c_i)$ for all i. Every finite subset of Σ together with these sentences has a model: a finite chain $c_1 < \cdots < c_n$ with the stated memberships is a finite frame, which has no infinite chain at all and therefore satisfies Σ. By compactness the whole set has a model, a frame in Σ's class with a D-free alternating chain, a contradiction. □

Remark 4.3. The analogy with common knowledge [4] is exact on this point: the common-knowledge operator is the greatest fixed point $\nu X.\, E(\varphi \wedge X)$, its induction axiom is valid on a frame exactly when the common-knowledge relation is the transitive closure of the union of the agents' relations, and that class is not first-order definable. Here $\bigcirc_C \vee \bigcirc_{C'}$ is the greatest fixed point of Proposition 2.3, the induction scheme (JL) is valid under the intersection reading exactly when the alternating chains are absent, and this class is not first-order definable.

5. Every nucleus is a join of relativized necessities

On a poset with an infinite ascending chain some nuclei on $\mathrm{Up}(P)$ are not relativized necessities [6, Thm. 2.12]; the join of the two nuclei of Example 3.3(a) is one. They are, however, joins of relativized necessities, and the generators can be written down.

For $a \in H$ the *closed* and *open* nuclei are $c_a(b) = a \vee b$ and $u_a(b) = a \to b$; they generate $\mathrm{N}(H)$ as a frame, and every nucleus is the join of the nuclei $u_a \wedge c_{ja}$, $a \in H$ [5, Ch. II, §2]. On $\mathrm{Up}(P)$ these generators are relativized necessities: for up-sets U, V,

$$\bigcirc_{U \cup (P \setminus V)} = u_U \wedge c_V,$$

since $\uparrow x \cap (U \cup (P \setminus V)) \subseteq W$ iff $\uparrow x \cap U \subseteq W$ and $\uparrow x \subseteq V \cup W$, i.e. $x \in (U \to W) \cap (V \cup W)$. In particular $u_U = \bigcirc_U$ and $c_V = \bigcirc_{P \setminus V}$. The following theorem is therefore the classical presentation in Alexandrov form, with a direct proof.

Theorem 5.1. *Let P be a poset and j a nucleus on $\mathrm{Up}(P)$. For $U \in \mathrm{Up}(P)$ and $x \in jU \setminus U$ put $C(U,x) = P \setminus (\uparrow x \setminus U) = U \cup (P \setminus \uparrow x)$, so that $\bigcirc_{C(U,x)} = u_U \wedge c_{\uparrow x}$. Then $\bigcirc_{C(U,x)} \le j$, $U \notin \mathrm{Fix}(\bigcirc_{C(U,x)})$, and*

$$j = \bigvee \{ \bigcirc_{C(U,x)} : \ U \in \mathrm{Up}(P),\ x \in jU \setminus U \} = \bigvee \{ \bigcirc_C : \ C \subseteq P,\ \bigcirc_C \le j \}.$$

Hence the relativized necessities are join-dense in $\mathrm{N}(\mathrm{Up}(P))$, and the sublattice they generate consists of the finite joins $\bigcirc_{C_1} \vee \cdots \vee \bigcirc_{C_n}$, each given by the greatest fixed point of Proposition 2.3 with n conditions.

Proof. Write $C = C(U,x)$. Since $\uparrow x \cap C \subseteq U$ we have $x \in \bigcirc_C U \setminus U$, so U is not fixed by $\bigcirc_C$. For $\bigcirc_C \le j$ we show $\mathrm{Fix}(j) \subseteq \mathrm{Fix}(\bigcirc_C)$. Let $V \in \mathrm{Fix}(j)$ and $z \notin V$, and suppose that no point of C above z lies outside V, i.e. $\uparrow z \setminus V \subseteq \uparrow x \setminus U$. Then z itself lies in $\uparrow x \setminus U$, so $z \ge x$ and therefore $z \in jU$; and $\uparrow z \cap U \subseteq V$, i.e. $z \in U \to V$, where $U \to V = \{w : \uparrow w \cap U \subseteq V\}$ is the Heyting implication of $\mathrm{Up}(P)$. As V is fixed by j, so is $U \to V$: since j preserves binary meets, $j(a \to b) \wedge ja = j((a \to b) \wedge a) \le jb$ gives $j(a \to b) \le ja \to jb$, whence $j(U \to V) \le jU \to jV = jU \to V \le U \to V$. Hence $z \in jU \cap j(U \to V) = j(U \cap (U \to V)) \subseteq jV = V$, a contradiction. So every $z \notin V$ has a point of $C \setminus V$ above it, $V \in \mathrm{Fix}(\bigcirc_C)$, and $\bigcirc_C \le j$.

Consequently $\mathrm{Fix}(j) \subseteq \mathrm{Fix}(\bigcirc_{C(U,x)})$ for all (U,x), with $U \notin \mathrm{Fix}(\bigcirc_{C(U,x)})$ for every $U \notin \mathrm{Fix}(j)$; so $\mathrm{Fix}(j) = \bigcap_{(U,x)} \mathrm{Fix}(\bigcirc_{C(U,x)})$, which is the first equality, and the second follows since every $\bigcirc_C \le j$ lies between $\bigcirc_{C(U,x)}$'s join and j. For the sublattice, meets of relativized necessities are relativized necessities (Lemma 2.2(ii)), and since $\mathrm{N}(H)$ is a frame [5], binary meets distribute over joins: $(\bigcirc_{C_1} \vee \cdots \vee \bigcirc_{C_n}) \wedge \bigcirc_E = \bigcirc_{C_1 \cup E} \vee \cdots \vee \bigcirc_{C_n \cup E}$. The greatest-fixed-point description of an n-fold join is proved as Proposition 2.3, with $X \subseteq {\downarrow}(X \cap C_i)$ required for every i. □

Remark 5.2. The generators $\bigcirc_{C(U,x)} = u_U \wedge c_{\uparrow x}$ refine the classical generators $u_U \wedge c_{jU}$: as $jU = \bigcup_{x \in jU} \uparrow x$ and closed nuclei turn joins into joins, $u_U \wedge c_{jU} = \bigvee_{x \in jU} (u_U \wedge c_{\uparrow x})$ by frame distributivity, and the terms with $x \in U$ are below the identity nucleus. What is specific to $\mathrm{Up}(P)$ is that these generators, and hence all finite meets of open and closed nuclei, are induced by subsets of P.

Example 5.3. Finite joins do not suffice. For any poset, $\bigcirc_C \leq \neg\neg$ if and only if C is cofinal, i.e. every point has a point of C above it: if $\bigcirc_C \leq \neg\neg$ then $\bigcirc_C \varnothing \subseteq \neg\neg\varnothing = \varnothing$, so no point has $\uparrow x \cap C = \varnothing$; conversely, if C is cofinal and $x \in \bigcirc_C U$, then every $y \geq x$ has a point $z \geq y$ in C, which lies in U, so $\uparrow y \cap U \neq \varnothing$ for all $y \geq x$, i.e. $x \in \neg\neg U$. Let T be the full binary tree, ordered by extension of finite 0–1 words. Given cofinal $C_1, \ldots, C_n$, choose $x_1 < x_2 < \cdots$ with $x_k \in C_i$ for $k \equiv i \pmod{n}$; then $B = {\downarrow}\{x_k\}$ is an infinite branch and $U = T \setminus B$ is an up-set fixed by every $\bigcirc_{C_i}$, each C_i being cofinal in B. But $\neg U = \{t : \uparrow t \cap U = \varnothing\}$ is empty: a node $t \notin B$ lies in U, and a node $t \in B$ has two children, of which the one not on B lies in U; so every node has a descendant in U, and $\neg\neg U = T \neq U$. Hence $\neg\neg$ on $\mathrm{Up}(T)$ is not a join of finitely many relativized necessities, while by Theorem 5.1 it is the join of all $\bigcirc_C$ with C cofinal.

Proposition 5.4. *Every nucleus on* $\mathrm{Up}(\omega)$ *is a join of at most two relativized necessities; for instance* $\neg\neg = \bigcirc_{\mathrm{even}} \vee \bigcirc_{\mathrm{odd}}$.

Proof. The up-sets of ω are $\varnothing$ and the tails $\uparrow n$, and $\mathrm{Fix}(\bigcirc_C)$ consists of ω, the tails $\uparrow(c+1)$ for $c \in C$, and $\varnothing$ exactly when C is infinite. The fixed-point set S of a nucleus contains ω and is closed under intersections, so if it contains infinitely many tails it contains $\varnothing$. Let $A = \{c : \uparrow(c+1) \in S\}$. If A is infinite, or if A is finite and $\varnothing \notin S$, then $S = \mathrm{Fix}(\bigcirc_A)$. If A is finite and $\varnothing \in S$, split $\omega \setminus A$ into two disjoint infinite sets E and O; then $\mathrm{Fix}(\bigcirc_{A\cup E}) \cap \mathrm{Fix}(\bigcirc_{A\cup O}) = S$, so the nucleus is $\bigcirc_{A\cup E} \vee \bigcirc_{A\cup O}$. For $\neg\neg$, $S = \{\varnothing, \omega\}$ and $A = \varnothing$. □

Question 5.5. Which nuclei on $\mathrm{Up}(P)$ are joins of finitely many relativized necessities? On ω two always suffice (Proposition 5.4); on the binary tree $\neg\neg$ needs infinitely many (Example 5.3).

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